吴越

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https://oftrongs.github.io

教育背景

清华大学，经济管理学院 管理科学与工程硕士 **北京，2024.9-2027.6**

- GPA：3.9/4.0 荣誉奖项：**清华-哥大商务分析双硕士项目**海外学习奖学金
- 研究方向：因果推断 核心课程：深度学习、高等机器学习、优化理论与方法、高等计量经济学、商务智能

哥伦比亚大学，工程学院 运筹学硕士 **纽约，2025.9-2026.5**

- 核心课程：高等强化学习、高维概率论、贝叶斯统计、数值模拟、分布式数据处理

中国人民大学，统计学院 学士，经济统计学 **北京，2020.9-2024.6**

- 核心课程：概率论、数理统计、回归分析、随机过程、多元统计分析、机器学习、程序设计、数据结构、数据库系统

实习经历

华为 2012 实验室，AI Infra 工程师实习生 **北京，2026.7-2026.9**

面向大模型推理加速场景，完成投机解码在多框架下的复现实测与分析，并基于 MindSpore Lite LLM 进行端侧功能支持与优化：

- 基于 L20 单卡完成 DFlash 多数据集 Transformers 框架下复现，分析投机解码相关指标如接受率分布、采样方式等；调研并复现 llama.cpp、MNN 支持投机推理调用方案，并验证量化版本加速效果，为端侧部署提供实验支撑
- 实现基于 Qwen3-4B 的 DFlash 投机解码，完成麒麟 NPU 的手机端侧推理，多提示词场景下实测达 4-6 倍无损加速（合作项目）
- 推理链路分析与性能优化，拆解各计算阶段如 Target verify、Drafting、Inject 等耗时与整体内存占用；定位 argmax 在 CPU 执行导致的 logits 回传瓶颈，优化推理流程中内存管理，减少单 token 约 608 KB 的数据拷贝，实测整体推理耗时降低 2%
- 对比块解码耗时为单解码约 1.1-1.2 倍；对于个别输出不一致问题，定位到两种解码的 logits 结果存在 10^{-4}-10^{-2} 数值精度差异

字节跳动，因果推断算法实习生 **北京，2025.2-2025.7**

面向电商优惠券投放场景，构建用户异质性因果效应深度模型，优化个性化发券策略以提升用户交易额和活跃度：

- 针对用户购买金额等连续型目标，根据长尾特征构建损失函数，并进行值域截断以达成稳定训练，相较 MSE 损失改善离线评测表现
- 面对多档有序优惠券干预任务，研究并实现 VCNet 框架，以及对应的多干预评测代码，在大促交易额业务上达成离线 EOM 收益
- 基于基线模型开展日活跃度场景的微调与候选模型筛选，定义离线分配特征指标预估策略风险，模型上线后日活数量较基线提升 1%
- 融合观测数据与随机实验数据，针对大规模观测性数据重叠性较差的情况，利用匹配法（Matching）将其与小规模随机对照实验数据进行融合，显著改善协变量平衡并提升异质处理效应估计的准确性，AUUC 由 0.79 提升至 0.86
- 分析输入变量与模型预测效应的相关性，识别与预测结果关联较强的特征，为 LHUC 等模型结构设计提供参考

项目经历

基于模仿学习和强化学习的自适应实验设计（课程项目） **2026.3-2026.5**

- 将多阶段实验资源分配问题建模为有限时域马尔可夫决策过程，设计和构建包含异质子群的仿真环境，支持多种子群效应分布；策略基于各子群当前估计特征与剩余预算，动态选择下一阶段采样子群，以降低最终整体估计误差
- 设计“贪心策略模仿学习+PPO 微调”算法，对比随机分配、贪心方差缩减等基线策略，显著提升了训练稳定性及最终策略表现

AI 对话智能体开发及用户行为分析（课程项目） **2025.4-2025.6**

- 基于 Coze 平台设计并搭建“模拟面签”对话智能体工作流，实现签证类型分流、多轮对话模拟、面签结果判定及用户交互数据自动回传等功能；设置严肃型和引导型两种签证官人格并完成随机分组 A/B 测试
- 对 58 条有效用户数据开展统计分析，引导型签证官通过率更高、严肃型签证官显著提升回答长度；提出隐式反馈收集等产品建议

结合外生变量信息的函数型数据对齐问题研究（本科论文） **2023.1-2024.5**

- 基于函数型数据对齐问题，探索估计平移量与元曲线的统计方法，在传统形状不变模型基础上，提出纳入外生变量信息的损失函数
- 利用 R 语言进行模拟实验，系统比较不同信噪比下的估计效果，发现当单条曲线观测效果欠佳时纳入外生变量信息能显著提升估计效果；将模型方法应用于 LOFAR 谱线检测问题，将谱线数据进行对齐并完成插值补全，有效还原谱图结构

目标变量样本分布不均的分类问题探索（课程项目） **2022.5-2022.6**

- 基于样本分布不均的二分类问题，探索 class_weight 的加权原理并将其应用于逻辑回归和决策树模型，发现对少类样本进行加权可以改善简单模型的分类效果；在不同模型中使用 Focal Loss 损失函数，发现在 XGBoost 等模型中改善效果明显

其他信息

- 个人主页：https://oftrongs.github.io
- 编程语言：Python (TensorFlow/PyTorch/Transformers/Gymnasium/Pyspark/scikit-learn), R, SQL
- 英语：IELTS 7.5, GRE 320, CET6
- 开源框架贡献：mindspore-lite/lite_llm（DFlash 板块），alibaba/MNN（16.1k stars，DFlash 冗余图问题修复 PR 已合并）

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EDUCATION

Tsinghua University, School of Economics and Management **Beijing, Sept 2024-Jun 2027**
Master of Management Science and Engineering

- Research Focus: Causal Inference Core Courses: Deep Learning, Optimization Theory, Advanced Econometrics

Columbia University, School of Engineering **New York, Sept 2025-May 2026**
Master of Science in Operations Research

- Core Courses: Advanced Reinforcement Learning, High-Dimensional Probability, Bayesian Statistics, Cloud Data Analytics

Renmin University of China, School of Statistics **Beijing, Sept 2020-Jun 2024**
Bachelor in Economic Statistics

- Core Courses: Regression Analysis, Stochastic Processes, Machine Learning, Data Structures, Database Systems

INTERNSHIP

Huawei 2012 Laboratories, AI Infrastructure Engineering Intern **Beijing, Jul 2026-Sept 2026**

- Reproduced and benchmarked DFlash across multiple datasets using Transformers; reproduced speculative decoding workflows in llama.cpp and MNN, and evaluated acceleration with quantized models to support on-device deployment
- Implemented DFlash for Qwen3-4B on Kirin NPU smartphones, achieving 4–6× speedup as part of a collaborative project
- Profiled inference stages and memory usage; identified logits transfers caused by CPU-side argmax and optimized memory management, reducing data copying by approximately 608 KB per token and end-to-end inference time by 2%
- Measured block decoding latency at approximately 1.1–1.2× single-token decoding latency; traced occasional output discrepancies to numerical differences of 10^{-4}–10^{-2} in logits between the two decoding modes

ByteDance, Causal Inference Algorithm Intern **Beijing, Feb 2025-Jul 2025**

- Developed deep learning models for heterogeneous treatment effects (uplift models) in e-commerce coupon allocation to optimize personalized coupon strategies and improve user spending and activity
- Designed loss functions tailored to long-tailed continuous outcomes and applied range clipping to stabilize training
- Investigated and implemented VCNet for multiple ordered coupon interventions, developed corresponding evaluation code, and achieved offline EOM gains in transaction-value optimization during major promotional campaigns
- Fine-tuned baseline models and selected candidate models for daily user activity optimization; defined offline allocation metrics to assess policy risk, with the deployed model increasing daily active users by 1% relative to the baseline
- Combined large-scale observational data with small-scale randomized controlled trial data using matching to address poor overlap, improving covariate balance and heterogeneous effect estimation, with AUUC increasing from 0.79 to 0.86

PROJECTS

Adaptive Experimental Design via Imitation Learning and Reinforcement Learning **Mar 2026-May 2026**

- Formulated multi-stage experimental resource allocation as a finite-horizon MDP; built a simulation environment with heterogeneous subgroups and multiple treatment-effect distributions, enabling adaptive subgroup selection based on current estimates and remaining budget to reduce final overall estimation error
- Developed a training pipeline combining imitation learning from a greedy variance reduction policy with PPO fine-tuning

AI Conversational Agent Development and User Behavior Analysis **Apr 2025-Jun 2025**

- Built a mock visa interview agent on Coze, supporting visa-type routing, multi-turn interviews and automated interaction data collection; conducted a randomized A/B test comparing strict and supportive interviewer personas
- Analyzed 58 valid user records: the supportive persona yielded higher pass rates, while the strict persona significantly increased response length; proposed product improvements, including implicit feedback collection

Functional Data Alignment with Exogenous Variables **Jan 2023-May 2024**

- Investigated functional data alignment and proposed a loss function incorporating exogenous variables
- Showed exogenous variables significantly improve estimation when individual curves were poorly observed; applied the method to LOFAR spectral line detection, aligning and interpolating spectral data to recover spectral structure

Classification with Imbalanced Target Classes **May 2022-Jun 2022**

- Investigated class weighting in imbalanced binary classification and applied it to logistic regression and decision trees, improving classification performance by upweighting minority-class samples
- Evaluated focal loss across different models, observing notable improvements in models such as XGBoost

ADDITIONAL INFORMATION

- Programming: Python (TensorFlow/PyTorch/Transformers/Gymnasium/PySpark/scikit-learn), R, SQL
- English: IELTS 7.5, GRE 320, CET-6
- Open-Source Contributions: DFlash-related contributions to mindspore-lite/lite_llm; merged PR fixing redundant graphs in the DFlash implementation of alibaba/MNN (16.1k stars)